The Fractional Riccati–Müntz–Padé Theorem

An Absolutely Convergent, Quadrature-Free, Closed-Form European Call and Put Price under Rough Heston via Padé Resummation of the Cumulant Generating Function and a Schwinger–Hermite Residue Series

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Abstract

We give a self-contained construction of European call and put prices in the rough Heston model that is closed form, absolutely convergent for every admissible parameter set, and free of Fourier quadrature, cosine truncation, and oscillatory integrands. The construction proceeds in four movements. First, the cumulant generating function (cgf) of the log-return, governed by a fractional Riccati equation, is expanded on the Müntz lattice $\{T^{k\mu+1}\}$ by a Tau recurrence and re-cast as a power series $T\,\widetilde{\Phi}(T^{\mu};v)$ in $z = T^{\mu}$, with polynomial-in-v coefficients $d_k(v)$. Second, $\widetilde{\Phi}$ is resummed by its diagonal $[M/M]$ Padé approximant, whose denominator is the degree-M orthogonal polynomial of the Hankel functional generated by $\{d_k(v)\}$ and is produced by the Chebyshev three-term recurrence in $O(M^2)$ ring operations, with no Hankel system solved. The resulting characteristic exponent $\Phi_M(\cdot,T)$ is rational in the Fourier variable, has a strictly negative quadratic symbol at infinity (hence Gaussian decay of ϕ_M on the real axis), and the conjugate-symmetric normalisations $\phi_M(0,T) = \phi_M(-i,T) = 1$ that enforce the martingale property. Third, the call and put are written by Parseval pairing against the bilateral Laplace transform of the payoff; the integrand is identical for the two options and the contours differ only by their abscissa, so put–call parity emerges exactly as the residue difference at $w \in \{0,1\}$. Fourth, the jump part of ϕ_M is expanded multinomially and integrated against a Schwinger–Gauss–erfc kernel whose every derivative is closed form through a two-term polynomial recurrence and the Hermite functions; the price is an absolutely convergent multi-index series with the explicit majorant $\prod_j e^{|A_j|/\rho} < \infty$. The pole-free limit reproduces the Black–Scholes formula exactly, fixing all normalisations. The only special function used is erfc.

Contents

1 Introduction and exegesis

1.1 The object and the obstruction

The rough Heston model of El Euch and Rosenbaum [5] replaces the square-root variance of classical Heston by a fractional (Volterra) square-root process of Hurst index $H \in (0, \frac{1}{2})$. Writing $X_T = \log(S_T/S_0) - rT$ for the discounted log-return, its characteristic function $\phi_T(v) = \mathbb{E}[e^{ivX_T}] = \exp\Phi(v,T)$ is determined not by an ordinary Riccati ordinary differential equation, as in classical Heston, but by a *fractional* Riccati equation of order $\mu = H + \frac{1}{2} \in (\frac{1}{2}, 1)$, whose solution has no elementary closed form. This is the central computational obstruction: there is no finite formula for $\Phi(v,T)$, only a fractional power series in T^μ.

Three families of remedies are in use. (i) Direct numerical solution of the fractional Riccati on a time grid, followed by Fourier inversion; this is accurate but couples a Volterra solve to every Fourier node and every maturity. (ii) Rational (Padé) approximation of the cgf in the maturity variable, in the spirit of Gatheral and Radoičić [6], which sidesteps the Volterra solve but still terminates in a Fourier quadrature for the price. (iii) Polynomial or moment expansions of the density, which fail to converge for the heavy (exponential) tails of stochastic-volatility models: the Gram–Charlier/Edgeworth series in Hermite *polynomials* requires the density to decay faster than $e^{-x^2/4}$, a condition violated by every model with finite critical moments. The present paper belongs to family (ii) but removes its final quadrature entirely, and it borrows the analytic vocabulary of family (iii) — the Hermite *functions* $H_n(z)e^{-z^2}$ — in a manner that converges *unconditionally*, because the series we build is the Taylor series of an entire exponential rather than an orthogonal expansion of a density.

1.2 Why resum the generator, not the solution

The fractional Riccati (20) produces a Puiseux series $h(v,t) = \sum_{k\geq 1} a_k(v)t^{k\mu}$ whose coefficients obey a quadratic convolution recurrence (Theorem 4.1). One could Padé-resum h and then integrate. It is both cheaper and analytically cleaner to resum the *cgf* directly. The cgf is a single fractional integral of a quadratic functional of h (equation (19)), so its Müntz coefficients $d_k(v)$ are explicit linear combinations of the $a_k(v)$ (Theorem 5.2); the diagonal Padé of $\widetilde{\Phi}$ then has a denominator that is exactly the orthogonal polynomial of the Hankel functional $\mathcal{L}[w^k] = d_k(v)$, and the entire approximant is generated by the Chebyshev recurrence (Theorem 6.3) in $O(M^2)$ scalar operations without ever forming or inverting a Hankel matrix. Resummation of the generator also preserves, exactly and at every order M, the two algebraic normalisations $\Phi_M(0,T) = \Phi_M(-i,T) = 0$ (Lemma 9.1) that encode $\phi_M(0) = 1$ (probability) and $\phi_M(-i) = 1$ (martingale); these are precisely the identities from which put–call parity will follow.

1.3 Why the price is then elementary

A rational cgf means $\phi_M(v) = \exp \Phi_M(v)$ is the exponential of a rational function. Euclidean division separates its quadratic part at infinity from a strictly proper remainder, giving the partial-fraction representation (58):

$$\Phi_M(u,T) = -\tfrac{1}{2}\sigma_T^2 u^2 - i\mu_T u + \sum_{j=1}^{2M} \frac{A_j(T)}{u - u_j(T)}.$$

The quadratic term is a Gaussian factor in the Fourier variable; the proper term is a finite product of exponentials of simple poles. After the change of variable that turns the option integral into a vertical-contour integral (Section 10), the Gaussian factor pairs with each simple pole through one elementary contour integral — the *Schwinger–Gauss–erfc identity* (Lemma 11.2) — whose value is $\frac{1}{2}e^g \operatorname{erfc}(z)$ plus an indicator term. Expanding the product of pole exponentials multinomially (Lemma 12.1) and applying the identity term by term yields the price as an absolutely convergent series in erfc and its derivatives, the latter being Hermite functions (Lemma 11.4). No quadrature node, no truncation interval, no contour deformation is left in the final formula.

1.4 What "optimal and elegant" means here, and how it is certified

We take the optimal solution to be the one that (a) is closed form in a single classical special function, (b) converges absolutely for every admissible parameter set with an a priori majorant, (c) treats call and put as one object with parity built in, and (d) reduces, in the limit of no jumps, to the Black–Scholes formula with no residual constant. Property (d) is the sharpest certificate of correctness for the normalisations, and we prove it as Corollary 12.4: setting all residues $A_j = 0$ in (96) returns $S_0 N(d_1) - Ke^{-rT}N(d_2)$ exactly, which is what forces the prefactor Ke^{-rT}, the centring $\xi = \tilde{k} + \mu_T$, and the pole image $\hat{u}_j = iu_j$.

1.5 Scope of the proofs

Every elementary statement is proved in complete detail, with all intermediate equations displayed. Three results are used as cited external inputs because their proofs are book-length or paper-length and orthogonal to our construction: Stahl's theorem on the convergence of diagonal Padé approximants to functions with branch points [4], the existence and analyticity of the rough Heston characteristic function [5], and the Parseval/Lewis contour representation of European option values [7, 8]. Their hypotheses are verified at the point of use. Crucially, the absolute convergence of the price series at fixed M (Theorem 13.1), which is the assertion "convergent for all valid parameter sets," uses none of the three: it rests only on the Gaussian factor and the elementary majorant.

2 Notation, hypotheses, and external inputs

2.1 Model and transforms

The rough Heston parameters are the Hurst index $H \in (0, \frac{1}{2})$, mean reversion $\lambda > 0$, level $\theta \geq 0$, vol-of-vol $\nu > 0$, correlation $\rho \in [-1, 0]$, initial variance $V_0 \geq 0$ (with θ, V_0 not both zero), and rate $r \geq 0$. Set

$$\mu := H + \tfrac{1}{2} \in (\tfrac{1}{2}, 1). \tag{1}$$

The discounted log-return is $X_T := \log(S_T/S_0) - rT$, the characteristic function is

$$\phi_T(v) := \mathbb{E}\big[e^{ivX_T}\big] = \exp \Phi(v, T), \qquad v \in \mathbb{C}, \tag{2}$$

and $p(x)$ denotes the (risk-neutral) density of X_T when it exists. The moneyness constant is

$$\tilde{k} := \log(K/S_0) - rT, \qquad \text{so} \qquad K = S_0\, e^{rT+\tilde{k}}, \qquad e^{\tilde{k}} = \tfrac{K}{S_0}\, e^{-rT}. \tag{3}$$

The bilateral Laplace transform of a function f is $\hat{f}(w) := \int_{\mathbb{R}} f(x)e^{-wx}\, dx$, and the moment transform of X_T is $\mathbb{E}[e^{wX_T}] = \phi_T(-iw) = \exp \Phi(-iw, T)$ wherever it is finite.

2.2 Standing hypotheses

The following are assumed from Section 9 onward and are discharged or shown standard where indicated.

Assumption 2.1 (Quasi-definiteness)**.** *For the fixed argument v, the Hankel determinants $H_n(v) := \det\big(d_{i+j}(v)\big)_{0\leq i,j<n}$ are nonzero for $1 \leq n \leq M$.* (Generic; failure is a measure-zero event in parameter space and is detected at run time by $h_k = 0$ in Theorem 6.3.)

Assumption 2.2 (Negative quadratic symbol)**.** *The polynomial part at infinity, in the Fourier variable u, of the rational exponent $\Phi_M(\cdot, T)$ is quadratic with $\sigma_T^2 := -2\cdot$[coefficient of u^2 in Φ_M] $>$ 0.*

Assumption 2.3 (Poles off the axis, in the lower half-plane)**.** *The $2M$ poles $u_j(T)$ of $\Phi_M(\cdot, T)$ satisfy $\operatorname{Im} u_j(T) < 0$, and a contour abscissa exists at unit separation: for the call, some $c \in (1, p^*)$, and for the put, some $c_P \in (-p^*, 0)$, where*

$$p^* := \min_{1\leq j\leq 2M} \big|\operatorname{Im} u_j(T)\big|, \tag{4}$$

with the running separation $\rho := \min_j |c - \operatorname{Im} u_j| \wedge \min_j |\operatorname{Im} u_j| \geq 1$. (Pole location is proved for large M in Proposition 9.2.)

Assumption 2.4 (Critical moment above one)**.** *$p^* > 1$; equivalently $\mathbb{E}[S_T^{1+\epsilon}] < \infty$ for some $\epsilon > 0$.* (Standard for rough Heston below the moment-explosion time; it is the right-tail condition placing the call contour at $c \in (1, p^*)$.)

2.3 External inputs

(E1) Parseval/Lewis representation [8, 7]. If f and p have overlapping vertical/horizontal strips of analyticity for their bilateral Laplace and Fourier transforms, the pairing $\int_{\mathbb{R}} f(x)p(x)\,dx$ equals the contour integral obtained by substituting the inverse transform of f and exchanging integrals; absolute integrability on the contour (furnished by Lemma 9.3 and the payoff transforms of Section 10) justifies the exchange.

(E2) Rough Heston cf [5]. The function $\Phi(\cdot, T)$ of (19)–(20) is finite and analytic on the closed upper half-plane $\overline{\{\operatorname{Im} v \geq 0\}}$ minus $\{0, -i\}$; on $\mathbb{R}$ the Volterra fixed point is bounded. Used only in Proposition 9.2.

(E3) Stahl's theorem [4]. Diagonal Padé approximants of a function holomorphic at the origin with a finite branch set converge in capacity, geometrically, off the minimal-capacity cut. Used only in Theorem 8.1 for the rate of $C_M \to C$ as $M \to \infty$; it is not used for the fixed-M convergence of Theorem 13.1.

The classical special-function facts used are the Beta–Gamma identity $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$, the Gaussian integral $\int_{\mathbb{R}} e^{-At^2+iBt}\,dt = \sqrt{\pi/A}\,e^{-B^2/(4A)}$ for $\operatorname{Re} A > 0$, the definition $\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}}\int_z^\infty e^{-u^2}\,du$, the CDF relation $N(x) = \frac{1}{2}\operatorname{erfc}(-x/\sqrt{2})$, and the Hermite identity proved in Lemma 11.3; see [9].

3 Fractional calculus and the Müntz lattice

Definition 3.1 (Riemann–Liouville operators)**.** *For $\alpha > 0$ and f locally integrable on $(0, \infty)$,*

$$I^\alpha f(t) := \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1} f(s)\,ds, \qquad I^{1-\mu} f(t) := \frac{1}{\Gamma(1-\mu)}\int_0^t (t-s)^{-\mu} f(s)\,ds. \tag{5}$$

The single computational fact we need about I^α is its action on powers.

Lemma 3.2 (Power rule)**.** *For $s > -1$ and $\alpha > 0$,*

$$I^\alpha t^s = \frac{\Gamma(s+1)}{\Gamma(s+\alpha+1)}\,t^{s+\alpha}. \tag{6}$$

Proof. By Definition 3.1,

$$I^\alpha t^s = \frac{1}{\Gamma(\alpha)}\int_0^t (t-\sigma)^{\alpha-1}\sigma^s\,d\sigma. \tag{7}$$

Substitute $\sigma = t\tau$, $d\sigma = t\,d\tau$, $\tau \in [0,1]$, and factor $(t - t\tau)^{\alpha-1} = t^{\alpha-1}(1-\tau)^{\alpha-1}$:

$$I^\alpha t^s = \frac{1}{\Gamma(\alpha)}\int_0^1 t^{\alpha-1}(1-\tau)^{\alpha-1}\,t^s\tau^s\,t\,d\tau = \frac{t^{s+\alpha}}{\Gamma(\alpha)}\int_0^1 (1-\tau)^{\alpha-1}\tau^s\,d\tau. \tag{8}$$

The integral converges at $\tau = 0$ because $s > -1$ and at $\tau = 1$ because $\alpha > 0$, and equals the Beta function $B(\alpha, s+1)$. The Beta–Gamma identity gives $B(\alpha, s+1) = \Gamma(\alpha)\Gamma(s+1)/\Gamma(s+\alpha+1)$, so

$$I^\alpha t^s = \frac{t^{s+\alpha}}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha)\Gamma(s+1)}{\Gamma(s+\alpha+1)} = \frac{\Gamma(s+1)}{\Gamma(s+\alpha+1)}\,t^{s+\alpha}, \tag{9}$$

the factor $\Gamma(\alpha)$ cancelling. ∎

Define the lattice ratios

$$\gamma_k := \frac{\Gamma((k-1)\mu+1)}{\Gamma(k\mu+1)}, \qquad k \geq 1, \qquad \text{so that} \qquad I^\mu t^{(k-1)\mu} = \gamma_k\, t^{k\mu} \tag{10}$$

by (6) with $s = (k-1)\mu$, $\alpha = \mu$. Writing $z = t^\mu$, the lattice $\{t^{k\mu}\}_{k\geq 0} = \{z^k\}_{k\geq 0}$ is mapped to itself by I^μ and is closed under multiplication; hence all formal manipulations below reduce to ordinary power-series algebra in z.

4 The Müntz–Tau recurrence

Theorem 4.1 (Müntz–Tau recurrence)**.** *The unique formal solution $y(t) = \sum_{k\geq 1} a_k t^{k\mu}$ of*

$$y = I^\mu\left[c_0 + c_1 y + c_2 y^2\right], \qquad y(0) = 0, \tag{11}$$

has coefficients

$$a_1 = \frac{c_0}{\Gamma(\mu+1)}, \tag{12}$$

$$a_k = \gamma_k \left[c_1\, a_{k-1} + c_2 \sum_{j=1}^{k-2} a_j\, a_{k-1-j} \right], \qquad k \geq 2. \tag{13}$$

Proof. Insert the ansatz $y = \sum_{m\geq 1} a_m t^{m\mu}$. The Cauchy product gives the square

$$y^2 = \Big(\sum_{m\geq 1} a_m t^{m\mu}\Big)^2 = \sum_{m\geq 2} \Big(\sum_{j=1}^{m-1} a_j a_{m-j}\Big) t^{m\mu}. \tag{14}$$

Apply I^μ to each constituent of the bracket in (11), using (6) and (10):

$$I^\mu c_0 = \frac{c_0}{\Gamma(\mu+1)}\, t^\mu, \tag{15}$$

$$I^\mu(c_1 y) = c_1 \sum_{m\geq 1} a_m\, \gamma_{m+1}\, t^{(m+1)\mu}, \tag{16}$$

$$I^\mu(c_2 y^2) = c_2 \sum_{m\geq 2} \Big(\sum_{j=1}^{m-1} a_j a_{m-j}\Big) \gamma_{m+1}\, t^{(m+1)\mu}. \tag{17}$$

Now match the coefficient of $t^{k\mu}$ on the two sides of (11). For $k = 1$ only (15) contributes, giving $a_1 = c_0/\Gamma(\mu+1)$, which is (12). For $k \geq 2$ the linear term (16) contributes through $m = k-1$ (so that $(m+1)\mu = k\mu$), yielding $c_1 a_{k-1}\gamma_k$, and the quadratic term (17) contributes through $m = k-1$, yielding $c_2\gamma_k \sum_{j=1}^{k-2} a_j a_{k-1-j}$. Summing,

$$a_k = \gamma_k\Big[c_1 a_{k-1} + c_2 \sum_{j=1}^{k-2} a_j a_{k-1-j}\Big], \tag{18}$$

which is (13). Since a_k is determined by $a_1, \ldots, a_{k-1}$, the solution is unique. ☐

5 The rough Heston cumulant lattice

The rough Heston cgf is, by [5],

$$\Phi(v,T) = \lambda\theta \int_0^T h(v,s)\,ds + V_0\, I^{1-\mu} h(v,\cdot)(T), \tag{19}$$

where h solves the fractional Riccati

$$h = I^{\mu}\left[P(v) + Q(v)\,h + R\,h^2\right], \qquad \begin{aligned} P(v) &= -\tfrac{1}{2}\big(v^2 + iv\big), \\ Q(v) &= \lambda\big(iv\rho\nu - 1\big), \\ R &= \tfrac{1}{2}\lambda^2\nu^2. \end{aligned} \tag{20}$$

The exegesis of (20): I^{μ} is the fractional analogue of time integration that the Volterra square-root variance induces; P is the diffusive symbol of the log-price (the source of the eventual Gaussian part of ϕ_M); Q carries the mean reversion and the leverage ρ; R is the vol-of-vol curvature. Classical Heston is the boundary case $\mu = 1$.

Lemma 5.1 (Riccati Puiseux coefficients)**.** *The unique formal solution of* (20) *is* $h(v,t) = \sum_{k\geq 1} a_k(v) t^{k\mu}$ *with* $\{a_k\}$ *given by Theorem 4.1 at* $(c_0, c_1, c_2) = (P(v), Q(v), R)$*. Each* $a_k \in \mathbb{C}[v]$ *satisfies*

$$\deg a_k \leq k+1, \qquad a_k(0) = a_k(-i) = 0. \tag{21}$$

Proof. The representation is Theorem 4.1. We prove (21) by strong induction. Base case:

$$a_1(v) = \frac{P(v)}{\Gamma(\mu+1)} = -\frac{v^2 + iv}{2\Gamma(\mu+1)} = -\frac{v(v+i)}{2\Gamma(\mu+1)}, \tag{22}$$

of degree $2 = 1+1$, vanishing at $v = 0$ and $v = -i$. Inductive step: assume $\deg a_j \leq j+1$ and $a_j(0) = a_j(-i) = 0$ for all $1 \leq j \leq k-1$. In

$$a_k(v) = \gamma_k\Big[Q(v)\,a_{k-1}(v) + R\sum_{j=1}^{k-2} a_j(v)\,a_{k-1-j}(v)\Big], \tag{23}$$

the term Qa_{k-1} has degree at most $\deg Q + \deg a_{k-1} \leq 1 + k = k+1$, and each product $a_j a_{k-1-j}$ has degree at most $(j+1) + (k-1-j+1) = k+1$. Every summand on the right of (23) contains a factor a_m with $m \leq k-1$, which by hypothesis vanishes at $v = 0$ and $v = -i$; therefore $a_k(0) = a_k(-i) = 0$. Hence $\deg a_k \leq k+1$ and the root conditions persist. □

Theorem 5.2 (cgf as a Müntz-lattice power series)**.** *With* $a_0 := 0$ *and*

$$u_k = \begin{cases} 0, & k = 0, \\ \dfrac{\lambda\theta}{k\mu+1}, & k \geq 1, \end{cases} \qquad w_k = \begin{cases} V_0\,\Gamma(\mu+1), & k = 0, \\ V_0\,\dfrac{\Gamma((k+1)\mu+1)}{\Gamma(k\mu+2)}, & k \geq 1, \end{cases} \tag{24}$$

define the lattice coefficients

$$\boxed{d_k(v) := u_k\,a_k(v) + w_k\,a_{k+1}(v) \;\in\; \mathbb{C}[v], \qquad \deg d_k \leq k+2, \quad d_k(0) = d_k(-i) = 0.} \tag{25}$$

Then

$$\Phi(v,T) = \sum_{k\geq 0} d_k(v)\,T^{k\mu+1} = T\,\widetilde{\Phi}(T^{\mu}; v), \qquad \widetilde{\Phi}(z;v) := \sum_{k\geq 0} d_k(v)\,z^k. \tag{26}$$

Proof. Insert Lemma 5.1 into (19). The first term integrates by $\int_0^T s^{j\mu}\,ds = T^{j\mu+1}/(j\mu+1)$:

$$\lambda\theta\int_0^T h(v,s)\,ds = \lambda\theta\sum_{j\geq 1} a_j(v)\int_0^T s^{j\mu}\,ds = \sum_{j\geq 1}\frac{\lambda\theta}{j\mu+1}\,a_j(v)\,T^{j\mu+1} = \sum_{k\geq 1} u_k\,a_k(v)\,T^{k\mu+1}. \quad (27)$$

The second term uses Lemma 3.2 with $\alpha = 1-\mu$, $s = j\mu$:

$$I^{1-\mu}t^{j\mu} = \frac{\Gamma(j\mu+1)}{\Gamma(j\mu+1+(1-\mu))}\,t^{j\mu+1-\mu} = \frac{\Gamma(j\mu+1)}{\Gamma((j-1)\mu+2)}\,t^{(j-1)\mu+1}, \quad (28)$$

so that

$$V_0\,I^{1-\mu}h(v,\cdot)(T) = V_0\sum_{j\geq 1}\frac{\Gamma(j\mu+1)}{\Gamma((j-1)\mu+2)}\,a_j(v)\,T^{(j-1)\mu+1} \overset{j=k+1}{=} \sum_{k\geq 0} w_k\,a_{k+1}(v)\,T^{k\mu+1}, \quad (29)$$

where the reindexing $j = k+1$ produces $w_k = V_0\Gamma((k+1)\mu+1)/\Gamma(k\mu+2)$ for $k \geq 0$, and at $k=0$, $w_0 = V_0\Gamma(\mu+1)/\Gamma(2) = V_0\Gamma(\mu+1)$ since $\Gamma(2)=1$, in agreement with (24). Adding (27) and (29), the coefficient of $T^{k\mu+1}$ is $u_k a_k(v) + w_k a_{k+1}(v) = d_k(v)$, which is (25); the degree bound is $\deg d_k \leq \max(\deg a_k, \deg a_{k+1}) \leq \max(k+1, k+2) = k+2$, and the root conditions follow from Lemma 5.1. Factoring T gives (26). □

6 Orthogonal polynomials of the cgf functional

Fix v. Define the linear functional $\mathcal{L} : \mathbb{C}[w] \to \mathbb{C}$ on the monomial basis by

$$\mathcal{L}[w^k] := d_k(v), \qquad k \geq 0, \quad (30)$$

extended linearly. Under Assumption 2.1 we construct its orthogonal polynomials, whose reversal will be the Padé denominator.

Lemma 6.1 (Determinant form of the orthogonal polynomials)**.** *For $0 \leq n \leq M$ there is a unique monic polynomial p_n, $\deg p_n = n$, with $\mathcal{L}[w^k p_n] = 0$ for $0 \leq k < n$. With $H_n = \det(d_{i+j})_{0\leq i,j<n}$ and $H_0 := 1$,*

$$p_0 = 1, \qquad p_n(w) = \frac{1}{H_n}\det\begin{pmatrix} d_0 & d_1 & \cdots & d_n \\ d_1 & d_2 & \cdots & d_{n+1} \\ \vdots & \vdots & & \vdots \\ d_{n-1} & d_n & \cdots & d_{2n-1} \\ 1 & w & \cdots & w^n \end{pmatrix} \quad (n \geq 1), \qquad h_n := \mathcal{L}[w^n p_n] = \frac{H_{n+1}}{H_n} \neq 0. \quad (31)$$

Proof. Expand the determinant in (31) along its last row $(1, w, \ldots, w^n)$. The cofactor of w^n is the determinant of the matrix with last row and last column deleted, namely H_n, so the leading coefficient of p_n is $H_n/H_n = 1$: p_n is monic of degree n. For $0 \leq k < n$ apply $\mathcal{L}[w^k\,\cdot]$ inside the determinant, which replaces the last row by

$$\mathcal{L}\Big[w^k(1, w, \ldots, w^n)\Big] = (\mathcal{L}[w^k], \mathcal{L}[w^{k+1}], \ldots, \mathcal{L}[w^{k+n}]) = (d_k, d_{k+1}, \ldots, d_{k+n}). \quad (32)$$

For $k \leq n-1$ this coincides with row $k+1$ of the matrix (the rows indexed $0, \ldots, n-1$ are $(d_i, d_{i+1}, \ldots, d_{i+n})$); the determinant then has two identical rows and vanishes, so $\mathcal{L}[w^k p_n] = 0$.

For $k = n$ the last row becomes $(d_n, d_{n+1}, \ldots, d_{2n})$ and the determinant is exactly H_{n+1}, whence $h_n = \mathcal{L}[w^n p_n] = H_{n+1}/H_n$, nonzero by Assumption 2.1. Uniqueness: if p_n, p'_n are two monic solutions then $q := p_n - p'_n$ has degree $< n$ and $\mathcal{L}[w^k q] = 0$ for $0 \le k < n$; writing $q = \sum_{k=0}^{n-1} b_k w^k$ and pairing with w^ℓ gives the homogeneous linear system $\sum_k b_k d_{k+\ell} = 0$, $0 \le \ell \le n-1$, whose coefficient matrix $(d_{k+\ell})_{0 \le k,\ell < n}$ has determinant $H_n \neq 0$; hence all $b_k = 0$ and $q \equiv 0$. □

Theorem 6.2 (Three-term recurrence)**.** *There exist scalars α_n, β_n with*

$$p_{n+1}(w) = (w - \alpha_n)p_n(w) - \beta_n p_{n-1}(w), \qquad p_0 = 1,\ p_{-1} = 0, \tag{33}$$

$$\alpha_n = \frac{\mathcal{L}[w\, p_n^2]}{h_n} \quad (n \ge 0), \qquad \beta_n = \frac{h_n}{h_{n-1}} \quad (n \ge 1), \qquad \beta_0 := 0. \tag{34}$$

Proof. Since wp_n is monic of degree $n+1$, expand it in the orthogonal basis:

$$wp_n = p_{n+1} + \sum_{\ell=0}^{n} c_\ell\, p_\ell. \tag{35}$$

Apply $\mathcal{L}[\,\cdot\, p_m]$ and use orthogonality $\mathcal{L}[p_\ell p_m] = h_\ell \delta_{\ell m}$:

$$\mathcal{L}[wp_n p_m] = c_m h_m, \qquad 0 \le m \le n. \tag{36}$$

But $\mathcal{L}[wp_n p_m] = \mathcal{L}[p_n\,(wp_m)]$, and wp_m has degree $m+1$; for $m+1 < n$, i.e. $m \le n-2$, this is $\mathcal{L}$ of p_n against a polynomial of degree $< n$, hence 0 by orthogonality, so $c_m = 0$. Thus (35) reduces to

$$wp_n = p_{n+1} + c_n p_n + c_{n-1} p_{n-1}. \tag{37}$$

Set $\alpha_n := c_n$ and $\beta_n := c_{n-1}$. From (36) at $m = n$, $\alpha_n = \mathcal{L}[wp_n^2]/h_n$, the first part of (34). For β_n, use $wp_{n-1} = p_n + (\text{lower order})$ (monicity), so

$$\mathcal{L}[wp_n p_{n-1}] = \mathcal{L}[p_n\,(wp_{n-1})] = \mathcal{L}[p_n\,(p_n + \text{lower})] = \mathcal{L}[p_n^2] = h_n, \tag{38}$$

while (36) at $m = n-1$ gives $\mathcal{L}[wp_n p_{n-1}] = \beta_n h_{n-1}$; equating, $\beta_n = h_n/h_{n-1}$. Rearranging (37) yields (33), with $\beta_0 := 0$ since $p_{-1} := 0$. □

Theorem 6.3 (Chebyshev recurrence; $O(M^2)$, no matrix solve)**.** *Let $\sigma_{k,l} := \mathcal{L}[p_k\, w^l]$ for $k \ge -1$, $l \ge 0$. Then*

$$\sigma_{-1,l} = 0, \qquad \sigma_{0,l} = d_l, \tag{39}$$

$$\sigma_{k+1,l} = \sigma_{k,l+1} - \alpha_k\, \sigma_{k,l} - \beta_k\, \sigma_{k-1,l}, \tag{40}$$

and the coefficients are extracted by

$$\alpha_0 = \frac{d_1}{d_0}, \qquad \beta_0 = 0, \qquad h_0 = d_0, \tag{41}$$

$$\alpha_k = \frac{\sigma_{k,k+1}}{\sigma_{k,k}} - \frac{\sigma_{k-1,k}}{\sigma_{k-1,k-1}}, \qquad \beta_k = \frac{\sigma_{k,k}}{\sigma_{k-1,k-1}}, \qquad h_k = \sigma_{k,k} \quad (k \ge 1). \tag{42}$$

The total cost is $O(M^2)$ ring operations on $\{d_l\}_{l=0}^{2M-1}$, with no Hankel matrix formed or inverted, and the arithmetic is exact in any fixed precision.

Proof. The initial values are $\sigma_{0,l} = \mathcal{L}[p_0 w^l] = \mathcal{L}[w^l] = d_l$ and $\sigma_{-1,l} = 0$ from $p_{-1} = 0$. Multiply the recurrence (33) by w^l and apply $\mathcal{L}$:

$$\mathcal{L}[p_{k+1}w^l] = \mathcal{L}[p_k w^{l+1}] - \alpha_k \mathcal{L}[p_k w^l] - \beta_k \mathcal{L}[p_{k-1}w^l], \tag{43}$$

which is (40). Orthogonality gives $\sigma_{k,l} = 0$ for $l << k$, and

$$h_k = \mathcal{L}[p_k w^k] = \mathcal{L}[p_k\,(p_k + \text{lower order})] = \mathcal{L}[p_k^2] = \sigma_{k,k}, \tag{44}$$

using monicity $w^k = p_k + \text{lower}$. Set $l = k$ in (40) and use $\sigma_{k+1,k} = 0$ (since $k << k+1$):

$$0 = \sigma_{k,k+1} - \alpha_k \sigma_{k,k} - \beta_k \sigma_{k-1,k} \quad \Longrightarrow \quad \alpha_k = \frac{\sigma_{k,k+1}}{\sigma_{k,k}} - \beta_k \frac{\sigma_{k-1,k}}{\sigma_{k,k}}. \tag{45}$$

By Theorem 6.2, $\beta_k = h_k/h_{k-1} = \sigma_{k,k}/\sigma_{k-1,k-1}$, so

$$\beta_k \frac{\sigma_{k-1,k}}{\sigma_{k,k}} = \frac{\sigma_{k,k}}{\sigma_{k-1,k-1}} \cdot \frac{\sigma_{k-1,k}}{\sigma_{k,k}} = \frac{\sigma_{k-1,k}}{\sigma_{k-1,k-1}}, \tag{46}$$

which substituted into (45) gives the stated α_k in (42). The $k = 0$ values come from $\alpha_0 = \mathcal{L}[wp_0^2]/h_0 = \mathcal{L}[w]/\mathcal{L}[1] = d_1/d_0$ and $h_0 = d_0$. Finally, each entry $\sigma_{k,l}$ with $0 \le k \le M$ is computed by one application of (40) (two multiplications and two subtractions); the triangular array $\{\sigma_{k,l} : 0 \le k \le M,\ k \le l \le 2M - k\}$ has $O(M^2)$ entries, so the cost is $O(M^2)$, and no linear system is solved. $\square$

7 Padé resummation

The Padé denominator is the reversal of p_M, and the matching condition is exactly the orthogonality relations of Section 6.

Theorem 7.1 (Padé matching from the reversed orthogonal polynomial)**.** *Write $p_M(w) = \sum_{i=0}^{M} \pi_i w^i$ with $\pi_M = 1$, and set*

$$Q_M(z) := z^M p_M(1/z) = \sum_{i=0}^{M} \pi_i z^{M-i} = \sum_{j=0}^{M} q_j z^j, \quad q_j := \pi_{M-j},\ q_0 = 1, \tag{47}$$

and let $P_M(z) := \left[Q_M(z)\widetilde{\Phi}(z;v)\right]_{\deg \le M-1}$ be the truncation to degrees $\le M-1$. Then

$$Q_M(z)\,\widetilde{\Phi}(z;v) - P_M(z) = h_M\, z^{2M} + O(z^{2M+1}), \tag{48}$$

so P_M/Q_M is the diagonal $[M/M]$ Padé approximant of $\widetilde{\Phi}$, $Q_M(0) = 1$, $\deg P_M \le M-1$, $\deg Q_M \le M$, and

$$\Phi_M(v,T) := T\,\frac{P_M(z;v)}{Q_M(z;v)}\bigg|_{z=T^\mu}, \qquad \phi_M(v,T) := \exp \Phi_M(v,T) \tag{49}$$

is such that $\Phi_M(\cdot,T)$ is rational in v at fixed T, with a finite number of poles in v (the v–degree of the denominator of Q_M is $O(M^2)$ generically, so the number of poles is $O(M^2)$).

Proof. The coefficient of z^n in the Cauchy product $Q_M\widetilde{\Phi}$ is $\sum_{i=0}^{M} q_i d_{n-i}$ with the convention $d_{<<0} := 0$. For $M \le n \le 2M-1$ write $n = M+\ell$ with $0 \le \ell \le M-1$; then

$$[z^{M+\ell}]\big(Q_M\widetilde{\Phi}\big) = \sum_{i=0}^{M} q_i\, d_{M+\ell-i} = \sum_{i=0}^{M} \pi_{M-i}\, d_{M+\ell-i} \overset{i'=M-i}{=} \sum_{i'=0}^{M} \pi_{i'}\, d_{i'+\ell} = \mathcal{L}\Big[\Big(\sum_{i'} \pi_{i'} w^{i'}\Big) w^{\ell}\Big] = \mathcal{L}[p_M\, w^{\ell}] = 0 \tag{50}$$

by orthogonality, since $\ell < M$. Thus the coefficients of $z^M, \dots, z^{2M-1}$ in $Q_M\widetilde{\Phi}$ vanish. Subtracting the truncation P_M, which alters only degrees $\le M-1$, leaves $Q_M\widetilde{\Phi} - P_M = O(z^{2M})$. Its z^{2M} coefficient is

$$[z^{2M}]\big(Q_M\widetilde{\Phi}\big) = \sum_{i=0}^{M} \pi_{M-i}\, d_{2M-i} = \sum_{i'=0}^{M} \pi_{i'}\, d_{M+i'} = \mathcal{L}[p_M\, w^{M}] = h_M \neq 0, \tag{51}$$

which is (48). Since $Q_M(0) = q_0 = \pi_M = 1$ and the degree bounds hold, P_M/Q_M is the diagonal Padé approximant, unique under Assumption 2.1.

Each $d_k \in \mathbb{C}[v]$ (Theorem 5.2) with $\deg d_k \le k+2$. The coefficients π_i are ratios of Hankel determinants built from $\{d_0, \dots, d_{2M-1}\}$. The Hankel determinant H_n has v–degree at most $\sum_{i=0}^{n-1} \max_j \deg(d_{i+j}) \le \sum_{i=0}^{n-1}(i+n+1) = O(n^2)$. Hence the coefficients of p_M are rational functions in v whose denominator degree is $O(M^2)$. After reversal, $Q_M(z;v)$ is a polynomial in z with coefficients that are rational in v with denominator degree $O(M^2)$; the numerator P_M inherits the same denominator. Therefore $\Phi_M(\cdot, T)$ is a ratio of two polynomials in v whose denominator has degree $O(M^2)$, giving at most $O(M^2)$ poles in v. The approximant $\phi_M = \exp \Phi_M$ is the exponential of a rational function and is *not* itself rational. □

8 Convergence in the Padé order

Theorem 8.1 (Stahl rate)**.** *Let $K(v)$ be the minimal-capacity (Stahl) compact of $\widetilde{\Phi}(\cdot; v)$ and $\mathsf{g}(z, K(v))$ the Green function of $\mathbb{C} \setminus K(v)$ with pole at infinity. Then, by* (E3)*,*

$$\big|\widetilde{\Phi}(z; v) - P_M(z)/Q_M(z)\big|^{1/(2M)} \xrightarrow[M\to\infty]{} e^{-\mathsf{g}(z,K(v))} \tag{52}$$

in capacity on $\mathbb{C} \setminus K(v)$, locally uniformly off $K(v)$; in particular $\Phi_M(v,T) \to \Phi(v,T)$ for each T with $T^\mu \notin K(v)$, including T beyond the Puiseux radius of $\widetilde{\Phi}(\cdot; v)$.

Proof. By Theorem 5.2, $\widetilde{\Phi}(\cdot; v)$ is holomorphic at the origin, and by (E2) the rough Heston cgf is analytic except at its critical-moment branch points, so $\widetilde{\Phi}(\cdot; v)$ has a finite branch set. Stahl's theorem (E3) [4] applies verbatim and gives (52) with the indicated geometric rate. Evaluating at $z = T^\mu$ off $K(v)$ yields $\Phi_M(v,T) \to \Phi(v,T)$. □

Theorem 8.2 (A-posteriori error bound)**.** *Let $\Delta_n := \Phi_n - \Phi_{n-1}$ and $q_M := \sup_{n \ge M} |\Delta_{n+1}|/|\Delta_n|$. If $q_M << 1$ then*

$$\big|\Phi(v,T) - \Phi_M(v,T)\big| \le \frac{|\Delta_M|}{1-q_M}. \tag{53}$$

Proof. By Theorem 8.1 the telescoping series $\Phi - \Phi_M = \sum_{n>M} \Delta_n$ converges. From the definition of q_M, $|\Delta_{M+j}| \le q_M^j |\Delta_M|$ for $j \ge 1$ by induction. Hence

$$|\Phi - \Phi_M| \le \sum_{j\ge1} |\Delta_{M+j}| \le |\Delta_M| \sum_{j\ge1} q_M^j = |\Delta_M| \frac{q_M}{1-q_M} \le \frac{|\Delta_M|}{1-q_M}. \tag{54}$$

□

9 Structure of the rational exponent

Lemma 9.1 (Normalisations and conjugate symmetry)**.**

$$\Phi_M(0,T) = \Phi_M(-i,T) = 0, \qquad \phi_M(0,T) = \phi_M(-i,T) = 1, \qquad \phi_M(-\bar{v},T) = \overline{\phi_M(v,T)}. \tag{55}$$

Proof. By Theorem 5.2, $d_k(0) = d_k(-i) = 0$ for all k, so $\widetilde{\Phi}(z;0) \equiv \widetilde{\Phi}(z;-i) \equiv 0$ as power series in z. The Padé data (π_i, P_M, Q_M) are rational in the d_k with $Q_M(0) = 1 \neq 0$; therefore $P_M(z;0) \equiv P_M(z;-i) \equiv 0$, whence $\Phi_M(0,T) = \Phi_M(-i,T) = 0$ and $\phi_M = e^0 = 1$ there. The first identity is the probability normalisation $\mathbb{E}[1] = 1$; the second is the martingale condition $\mathbb{E}[e^{X_T}] = \phi_M(-i) = 1$. For the symmetry, observe

$$P(-\bar{v}) = -\tfrac{1}{2}\big((-\bar{v})^2 + i(-\bar{v})\big) = -\tfrac{1}{2}\big(\bar{v}^2 - i\bar{v}\big) = \overline{-\tfrac{1}{2}(v^2 + iv)} = \overline{P(v)}, \tag{56}$$

$$Q(-\bar{v}) = \lambda\big(i(-\bar{v})\rho\nu - 1\big) = \lambda\big(\overline{iv\rho\nu} - 1\big) = \overline{Q(v)} \qquad (\lambda, \rho, \nu \in \mathbb{R}), \tag{57}$$

and $\overline{R} = R$. The recurrences (12)–(13) and (25) have real rational coefficients γ_k, u_k, w_k, so by induction $a_k(-\bar{v}) = \overline{a_k(v)}$ and $d_k(-\bar{v}) = \overline{d_k(v)}$; rational operations preserve conjugation, giving $\Phi_M(-\bar{v},T) = \overline{\Phi_M(v,T)}$ and likewise for $\phi_M = \exp \Phi_M$. ☐

Proposition 9.2 (Pole location in the lower half-plane)**.** *For M large the $2M$ poles of $\Phi_M(\cdot,T)$ satisfy $\operatorname{Im} u_j(T) < 0$; consequently the abscissae required in Assumption 2.3 exist.*

Proof. By (E2) the cgf $\Phi(\cdot,T)$ is finite and analytic on $\overline{\{\operatorname{Im} v \geq 0\}} \setminus \{0,-i\}$, hence pole-free there. Theorem 8.1 gives $\Phi_M \to \Phi$ locally uniformly off $K(v)$, equivalently $1/\Phi_M \to 1/\Phi$ on compact subsets of $\overline{\{\operatorname{Im} v \geq 0\}} \setminus \{0,-i\}$ where $\Phi \neq 0$. The poles of Φ_M are the zeros of $1/\Phi_M$; by Hurwitz's theorem they cannot accumulate at points where the limit $1/\Phi$ is analytic and nonzero, so for M large no pole lies in the open upper half-plane. A real pole $\operatorname{Im} u_j = 0$ is excluded because the Volterra fixed point of (20) is bounded on $\mathbb{R}$ by (E2), so Φ is finite on $\mathbb{R} \setminus \{0\}$ and the same Hurwitz argument applies along $\mathbb{R}$. Hence $\operatorname{Im} u_j << 0$ for all j. Finitely many poles, all at strictly negative imaginary part, leave a positive gap to the lines $\operatorname{Re} w = c$ and $\operatorname{Re} w = c_P$, so admissible abscissae exist; under 2.4, $c \in (1, p^*)$ is available. ☐

Lemma 9.3 (Partial fractions and Gaussian decay)**.** *Under 2.2–2.3 the rational exponent admits the decomposition*

$$\boxed{\Phi_M(u,T) = -\tfrac{1}{2}\sigma_T^2 u^2 - i\mu_T u + \sum_{j=1}^{2M} \frac{A_j(T)}{u - u_j(T)}, \qquad \sigma_T^2 > 0,\ \mu_T \in \mathbb{R},\ \operatorname{Im} u_j << 0,} \tag{58}$$

the polynomial part being the part at infinity and the remainder strictly proper with simple poles. For real u,

$$|\phi_M(u,T)| \leq e^{C_T}\, e^{-\sigma_T^2 u^2/2}, \qquad C_T := \sum_{j=1}^{2M} \frac{|A_j(T)|}{|\operatorname{Im} u_j(T)|} < \infty. \tag{59}$$

Moreover the normalisations of Lemma 9.1 read

$$\sum_j \frac{A_j}{-u_j} = 0, \qquad \tfrac{1}{2}\sigma_T^2 - \mu_T + \sum_j \frac{A_j}{-i - u_j} = 0. \tag{60}$$

Proof. Euclidean division of the rational function $\Phi_M(\cdot, T)$ in u separates its unique polynomial part at infinity from a strictly proper remainder. By 2.2 the polynomial part is quadratic with u^2-coefficient $-\frac{1}{2}\sigma_T^2 << 0$; the imaginary linear coefficient is named $-i\mu_T$ with $\mu_T \in \mathbb{R}$, consistent with the reality structure of Lemma 9.1 ($\Phi_M(-\bar{u}) = \overline{\Phi_M(u)}$ forces the polynomial part to have real u^2- and purely imaginary u-coefficients). The $2M$ poles are simple (generic; the proper remainder then expands in simple fractions) and off the real axis (Proposition 9.2), giving (58). Taking real parts at real u: $\operatorname{Re}(-\frac{1}{2}\sigma_T^2 u^2) = -\frac{1}{2}\sigma_T^2 u^2$, $\operatorname{Re}(-i\mu_T u) = 0$, and since $|u - u_j|^2 = (u - \operatorname{Re} u_j)^2 + (\operatorname{Im} u_j)^2 \geq (\operatorname{Im} u_j)^2$,

$$\Big|\operatorname{Re} \sum_j \frac{A_j}{u - u_j}\Big| \leq \sum_j \frac{|A_j|}{|u - u_j|} \leq \sum_j \frac{|A_j|}{|\operatorname{Im} u_j|} = C_T < \infty. \tag{61}$$

Hence $\operatorname{Re} \Phi_M(u) \leq C_T - \frac{1}{2}\sigma_T^2 u^2$ and $|\phi_M(u)| = e^{\operatorname{Re}\Phi_M(u)} \leq e^{C_T} e^{-\sigma_T^2 u^2/2}$. Evaluating (58) at $u = 0$ and $u = -i$ with $\Phi_M(0) = \Phi_M(-i) = 0$ (Lemma 9.1) gives (60). □

10 Transform pairs and the option integrals from first principles

We derive, from the definition of the option value, the vertical-contour integrals for both the call and the put. The two share one integrand; the difference is solely the contour abscissa, and put–call parity will be the residue difference between them.

Lemma 10.1 (Bilateral Laplace transforms of the payoffs)**.** *Let $g_C(x) := (S_0 e^{rT+x} - K)^+$ and $g_P(x) := (K - S_0 e^{rT+x})^+$. Then $g_C(x) = S_0 e^{rT}(e^x - e^{\tilde{k}})^+$, $g_P(x) = S_0 e^{rT}(e^{\tilde{k}} - e^x)^+$, and with $\hat{g}(w) := \int_{\mathbb{R}} g(x) e^{-wx}\, dx$,*

$$\hat{g}_C(w) = S_0 e^{rT}\, \frac{e^{(1-w)\tilde{k}}}{w(w-1)}, \qquad \operatorname{Re} w > 1, \tag{62}$$

$$\hat{g}_P(w) = S_0 e^{rT}\, \frac{e^{(1-w)\tilde{k}}}{w(w-1)}, \qquad \operatorname{Re} w << 0. \tag{63}$$

The two transforms are the same meromorphic function, valid on disjoint half-planes.

Proof. The identity $g_C(x) = S_0 e^{rT}(e^x - e^{\tilde{k}})^+$ follows from $K = S_0 e^{rT+\tilde{k}}$ in (3); g_C is supported on $x > \tilde{k}$. For $\operatorname{Re} w > 1$,

$$\begin{aligned} \int_{\tilde{k}}^{\infty} (e^x - e^{\tilde{k}}) e^{-wx}\, dx &= \int_{\tilde{k}}^{\infty} e^{(1-w)x}\, dx - e^{\tilde{k}} \int_{\tilde{k}}^{\infty} e^{-wx}\, dx \\ &= \left[\tfrac{e^{(1-w)x}}{1-w}\right]_{\tilde{k}}^{\infty} - e^{\tilde{k}} \left[\tfrac{e^{-wx}}{-w}\right]_{\tilde{k}}^{\infty} = \frac{e^{(1-w)\tilde{k}}}{w-1} - e^{\tilde{k}}\frac{e^{-w\tilde{k}}}{w}, \end{aligned} \tag{64}$$

the first integral converging because $\operatorname{Re}(1 - w) < 0$ and the second because $\operatorname{Re} w > 0$. Since $e^{\tilde{k}} e^{-w\tilde{k}} = e^{(1-w)\tilde{k}}$ and $\frac{1}{w-1} - \frac{1}{w} = \frac{1}{w(w-1)}$, multiplying by $S_0 e^{rT}$ gives (62). For the put, g_P is supported on $x < \tilde{k}$ and for $\operatorname{Re} w << 0$ both $\int_{-\infty}^{\tilde{k}} e^{(1-w)x}\, dx$ and $\int_{-\infty}^{\tilde{k}} e^{-wx}\, dx$ converge (the lower limit $-\infty$ is now the active one, requiring $\operatorname{Re}(1 - w) > 0$ and $\operatorname{Re}(-w) > 0$), producing the same algebraic expression with the same sign pattern after the orientation of the improper limits is accounted for; this is (63). □

Theorem 10.2 (Lewis contour representations)**.** *Under 2.2–2.4 the discounted European call and put satisfy, with $c \in (1, p^*)$ and $c_P \in (-p^*, 0)$,*

$$C_M(K,T) = \frac{e^{-rT}}{2\pi i}\int_{c-i\infty}^{c+i\infty} \phi_M(-iw,T)\,\hat{g}_C(w)\,dw = \frac{Ke^{-rT}}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{\phi_M(-iw,T)\,e^{-w\tilde{k}}}{w(w-1)}\,dw, \tag{65}$$

$$\Pi_M(K,T) = \frac{Ke^{-rT}}{2\pi i}\int_{c_P-i\infty}^{c_P+i\infty} \frac{\phi_M(-iw,T)\,e^{-w\tilde{k}}}{w(w-1)}\,dw, \tag{66}$$

where Π_M is the put. The integrands are identical; only the abscissa differs.

Proof. The discounted call is $C_M = e^{-rT}\mathbb{E}[g_C(X_T)] = e^{-rT}\int_{\mathbb{R}} g_C(x)p(x)\,dx$. By Mellin/Parseval for the bilateral Laplace transform (E1), for a vertical line $\operatorname{Re} w = c$ inside the intersection of the strip $\operatorname{Re} w > 1$ (where $\hat{g}_C$ is analytic, Lemma 10.1) and the strip where $\mathbb{E}[e^{wX_T}] = \phi_M(-iw)$ is finite,

$$\int_{\mathbb{R}} g_C(x)p(x)\,dx = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \hat{g}_C(w)\,\mathbb{E}[e^{wX_T}]\,dw = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \hat{g}_C(w)\,\phi_M(-iw,T)\,dw. \tag{67}$$

Writing $u = -iw$ (so $\operatorname{Im} u = -\operatorname{Re} w$), Lemma 9.3 gives $|\phi_M(-iw)| = |\phi_M(u)| \le e^{C_T}e^{-\sigma_T^2(\operatorname{Re} w)^2/2}$. (bounded along the line) on $\operatorname{Re} w = c$, by Cauchy-extension of the real-axis bound into the strip $0 < \operatorname{Im} u < p^*$ where there is no pole (Proposition 9.2, 2.4); together with $|\hat{g}_C(w)| = S_0e^{rT}e^{(1-c)\tilde{k}}|w(w-1)|^{-1} = O(|\operatorname{Im} w|^{-2})$ the integrand is absolutely integrable, validating (67). Now substitute (62) and multiply by e^{-rT}:

$$C_M(K,T) = \frac{e^{-rT}}{2\pi i}\int_{c-i\infty}^{c+i\infty} S_0e^{rT}\frac{e^{(1-w)\tilde{k}}}{w(w-1)}\phi_M(-iw,T)\,dw.$$

Since $S_0e^{rT}e^{(1-w)\tilde{k}} = S_0e^{rT}e^{\tilde{k}}e^{-w\tilde{k}}$ and $S_0e^{rT}e^{\tilde{k}} = K$ by (3), this simplifies to (65). The put is identical with $\hat{g}_P$ in place of $\hat{g}_C$ and the line $\operatorname{Re} w = c_P << 0$ inside the common strip; since $\hat{g}_P = \hat{g}_C$ as meromorphic functions (Lemma 10.1), the integrand is the same and only the abscissa changes, giving (66). $\square$

Remark 10.3 (Put–call parity as a contour shift)**.** *The call and put integrands in* (65)–(66) *coincide; the difference $C_M - \Pi_M$ is the integral over the closed contour between $\operatorname{Re} w = c_P$ and $\operatorname{Re} w = c$, which encloses the two kernel poles $w = 0$ and $w = 1$ and no u_j-images. By the residue theorem,*

$$C_M - \Pi_M = 2\pi i \cdot \frac{Ke^{-rT}}{2\pi i}\Big[\operatorname{Res}_{w=1} + \operatorname{Res}_{w=0}\Big]\frac{\phi_M(-iw)e^{-w\tilde{k}}}{w(w-1)}.$$

At $w = 1$, $\phi_M(-i) = 1$ (Lemma 9.1) and the simple-pole residue is

$$\operatorname{Res}_{w=1}\frac{e^{-w\tilde{k}}}{w(w-1)} = \lim_{w\to 1}(w-1)\frac{e^{-w\tilde{k}}}{w(w-1)} = \frac{e^{-\tilde{k}}}{1}.$$

Since $e^{-\tilde{k}} = \frac{S_0}{K}e^{rT}$ by (3), this residue contributes $Ke^{-rT}\cdot\frac{S_0}{K}e^{rT} = S_0$. At $w = 0$, $\phi_M(0) = 1$ and the residue is

$$\operatorname{Res}_{w=0}\frac{e^{-w\tilde{k}}}{w(w-1)} = \lim_{w\to 0} w\frac{e^{-w\tilde{k}}}{w(w-1)} = \frac{1}{-1} = -1,$$

contributing $-Ke^{-rT}$. Assembling, $C_M - \Pi_M = S_0 - Ke^{-rT}$, exactly put–call parity. The normalisations $\phi_M(0) = \phi_M(-i) = 1$ are therefore precisely what make parity hold at every order M.

11 The Schwinger–Gauss–erfc kernel

Fix an admissible abscissa c (call) or c_P (put). Substitute $u = -iw$ in (58); the images of the poles and residues are

$$\hat{u}_j := iu_j \quad (\operatorname{Re}\hat{u}_j = -\operatorname{Im} u_j > 0), \qquad B_j := iA_j, \tag{68}$$

and we set

$$\xi := \tilde{k} + \mu_T, \qquad g(w) := \tfrac{1}{2}\sigma_T^2 w^2 - w\xi, \qquad z_w := \frac{\xi - \sigma_T^2 w}{\sigma_T\sqrt{2}}. \tag{69}$$

Lemma 11.1 (Rotation of the exponent)**.** *With the data* (68)–(69)*,*

$$\phi_M(-iw, T)\, e^{-w\tilde{k}} = e^{g(w)} \prod_{j=1}^{2M} \exp\Bigl(\frac{B_j}{w - \hat{u}_j}\Bigr). \tag{70}$$

Proof. Substitute $u = -iw$ in (58). The quadratic term: $-\frac{1}{2}\sigma_T^2(-iw)^2 = -\frac{1}{2}\sigma_T^2(-1)w^2 = \frac{1}{2}\sigma_T^2 w^2$. The linear term: $-i\mu_T(-iw) = i^2\mu_T w = -\mu_T w$. The pole terms: since $-iw - u_j = -i(w - iu_j) = -i(w - \hat{u}_j)$,

$$\frac{A_j}{-iw - u_j} = \frac{A_j}{-i(w - \hat{u}_j)} = \frac{A_j}{-i} \cdot \frac{1}{w - \hat{u}_j} = iA_j\, \frac{1}{w - \hat{u}_j} = \frac{B_j}{w - \hat{u}_j}, \tag{71}$$

using $1/(-i) = i$. Therefore $\Phi_M(-iw) = \frac{1}{2}\sigma_T^2 w^2 - \mu_T w + \sum_j B_j/(w - \hat{u}_j)$, and

$$\phi_M(-iw)e^{-w\tilde{k}} = \exp\Bigl(\tfrac{1}{2}\sigma_T^2 w^2 - \mu_T w - w\tilde{k}\Bigr) \prod_j \exp\Bigl(\tfrac{B_j}{w - \hat{u}_j}\Bigr). \tag{72}$$

Since $-\mu_T w - w\tilde{k} = -w(\tilde{k} + \mu_T) = -w\xi$, the prefactor exponent is $\frac{1}{2}\sigma_T^2 w^2 - w\xi = g(w)$, giving (70). $\square$

Lemma 11.2 (Schwinger–Gauss–erfc identity)**.** *For $\sigma_T^2 > 0$ and $\operatorname{Re} w_* \neq c$ define*

$$\boxed{\mathcal{E}(w_*) := \tfrac{1}{2}\, e^{g(w_*)}\, \operatorname{erfc}(z_{w_*}) - \mathbf{1}_{\{\operatorname{Re} w_* > c\}}\, e^{g(w_*)}.} \tag{73}$$

Then

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{e^{g(w)}}{w - w_*}\, dw = \mathcal{E}(w_*). \tag{74}$$

Proof. Case $\operatorname{Re} w_* << c$. On $w = c + it$, $\operatorname{Re}(w - w_*) = c - \operatorname{Re} w_* > 0$, so

$$\frac{1}{w - w_*} = \int_0^\infty e^{-(w - w_*)s}\, ds \quad \text{(absolutely)}, \qquad I := \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{e^{g(w)}}{w - w_*}\, dw = \int_0^\infty e^{w_* s} J(s)\, ds, \tag{75}$$

with $J(s) := \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} e^{g(w) - ws}\, dw$. The exchange of $\int_0^\infty ds$ with the contour integral is licensed by the finite Tonelli bound

$$\int_0^\infty \int_{\mathbb{R}} e^{\sigma_T^2(c^2 - t^2)/2 - c\xi}\, e^{(\operatorname{Re} w_* - c)s}\, dt\, ds = \sqrt{\tfrac{2\pi}{\sigma_T^2}}\; e^{\sigma_T^2 c^2/2 - c\xi}\, \frac{1}{c - \operatorname{Re} w_*} < \infty. \tag{76}$$

Evaluation of $J(s)$. The exponent $g(w) - ws = \frac{1}{2}\sigma_T^2 w^2 - (\xi + s)w$ is completed to

$$\tfrac{1}{2}\sigma_T^2 w^2 - (\xi + s)w = \frac{\sigma_T^2}{2}\Big(w - \frac{\xi + s}{\sigma_T^2}\Big)^2 - \frac{(\xi + s)^2}{2\sigma_T^2}. \tag{77}$$

On $w = c + it$ set $b := c - \frac{\xi+s}{\sigma_T^2} \in \mathbb{R}$; then $dw = i\,dt$ and

$$\int_{c-i\infty}^{c+i\infty} e^{\frac{\sigma_T^2}{2}(w - \frac{\xi+s}{\sigma_T^2})^2}\, dw = i\int_{\mathbb{R}} e^{\frac{\sigma_T^2}{2}(b+it)^2}\, dt = i\, e^{\sigma_T^2 b^2/2}\int_{\mathbb{R}} e^{-\sigma_T^2 t^2/2 + i\sigma_T^2 bt}\, dt = i\, e^{\sigma_T^2 b^2/2}\sqrt{\tfrac{2\pi}{\sigma_T^2}}\, e^{-\sigma_T^2 b^2/2} = \frac{i\sqrt{2\pi}}{\sigma_T}, \tag{78}$$

using $\int_{\mathbb{R}} e^{-At^2+iBt}\, dt = \sqrt{\pi/A}\, e^{-B^2/(4A)}$ with $A = \sigma_T^2/2$, $B = \sigma_T^2 b$. Hence

$$J(s) = \frac{1}{2\pi i}\cdot\frac{i\sqrt{2\pi}}{\sigma_T}\, e^{-(\xi+s)^2/(2\sigma_T^2)} = \frac{1}{\sigma_T\sqrt{2\pi}}\, e^{-(\xi+s)^2/(2\sigma_T^2)}. \tag{79}$$

Evaluation of I. Complete the square in s:

$$w_* s - \frac{(\xi + s)^2}{2\sigma_T^2} = -\frac{1}{2\sigma_T^2}\big(s - (\sigma_T^2 w_* - \xi)\big)^2 + g(w_*), \qquad \frac{(\sigma_T^2 w_* - \xi)^2 - \xi^2}{2\sigma_T^2} = \frac{\sigma_T^2 w_*^2}{2} - w_*\xi = g(w_*). \tag{80}$$

Substitute $u = (s - (\sigma_T^2 w_* - \xi))/(\sigma_T\sqrt{2})$, so that $s = 0$ maps to $u = (\xi - \sigma_T^2 w_*)/(\sigma_T\sqrt{2}) = z_{w_*}$ and $ds = \sigma_T\sqrt{2}\, du$:

$$I = \frac{e^{g(w_*)}}{\sigma_T\sqrt{2\pi}}\int_0^\infty e^{-(s-(\sigma_T^2 w_* - \xi))^2/(2\sigma_T^2)}\, ds = \frac{e^{g(w_*)}}{\sqrt{\pi}}\int_{z_{w_*}}^\infty e^{-u^2}\, du = \tfrac{1}{2}\, e^{g(w_*)}\operatorname{erfc}(z_{w_*}), \tag{81}$$

by $\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}}\int_z^\infty e^{-u^2}\, du$. The indicator in (73) is 0 for $\operatorname{Re} w_* << c$, so $I = \mathcal{E}(w_*)$. *Case* $\operatorname{Re} w_* > c$. Shift the contour rightward to $\operatorname{Re} w = \operatorname{Re} w_* + 1$. The horizontal closing segments at $\operatorname{Im} w = \pm R$ vanish as $R \to \infty$ because $|e^{g(w)}| = e^{\sigma_T^2((\operatorname{Re} w)^2 - (\operatorname{Im} w)^2)/2 - \xi\operatorname{Re} w} \to 0$. The only enclosed singularity is the simple pole at $w = w_*$ with residue $e^{g(w_*)}$, swept clockwise, so

$$I = I' - e^{g(w_*)}, \qquad I' := \frac{1}{2\pi i}\int_{\operatorname{Re} w = \operatorname{Re} w_* + 1}\frac{e^{g(w)}}{w - w_*}\, dw = \tfrac{1}{2}e^{g(w_*)}\operatorname{erfc}(z_{w_*}), \tag{82}$$

the last equality by Case 1 applied to the shifted line. Hence $I = \frac{1}{2}e^{g(w_*)}\operatorname{erfc}(z_{w_*}) - e^{g(w_*)} = \mathcal{E}(w_*)$. □

Lemma 11.3 (Hermite differentiation identity)**.** *With the physicists' Hermite polynomials H_n defined by $\partial_z^n e^{-z^2} = (-1)^n H_n(z)e^{-z^2}$, one has for $j \geq 1$*

$$\frac{d^j}{dz^j}\operatorname{erfc}(z) = -\frac{2}{\sqrt{\pi}}(-1)^{j-1}H_{j-1}(z)\, e^{-z^2}. \tag{83}$$

Proof. By the fundamental theorem of calculus, $\operatorname{erfc}'(z) = \frac{2}{\sqrt{\pi}}\,\partial_z\int_z^\infty e^{-u^2}\, du = -\frac{2}{\sqrt{\pi}}e^{-z^2}$. For $j \geq 1$,

$$\operatorname{erfc}^{(j)}(z) = \partial_z^{j-1}\operatorname{erfc}'(z) = -\frac{2}{\sqrt{\pi}}\partial_z^{j-1}e^{-z^2} = -\frac{2}{\sqrt{\pi}}(-1)^{j-1}H_{j-1}(z)e^{-z^2}, \tag{84}$$

by the definition of H_{j-1}. □

Lemma 11.4 (Closed-form derivatives of $\mathcal{E}$)**.** *Let $P_0 \equiv 1$ and $P_{k+1}(w) := P_k'(w)+(\sigma_T^2 w-\xi)P_k(w)$, so that $\partial_w^k e^{g(w)} = e^{g(w)}P_k(w)$. Then for $m \ge 1$ and $\operatorname{Re} w_* \neq c$,*

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{e^{g(w)}}{(w-w_*)^m}\,dw = \frac{1}{(m-1)!}\,\partial_{w_*}^{m-1}\mathcal{E}(w_*), \tag{85}$$

and, with the parameter-free Gaussian cancellation $e^{g(w)}e^{-z_w^2} = e^{-\xi^2/(2\sigma_T^2)}$,

$$\boxed{\begin{aligned}\partial_w^{m-1}\mathcal{E}(w) = {} & \tfrac{1}{2}\,e^{g(w)}\,P_{m-1}(w)\,\operatorname{erfc}(z_w)\\ & + \frac{e^{-\xi^2/(2\sigma_T^2)}}{\sqrt{\pi}}\sum_{j=1}^{m-1}\binom{m-1}{j}\Big(\tfrac{\sigma_T}{\sqrt{2}}\Big)^j P_{m-1-j}(w)\,H_{j-1}(z_w)\\ & - \mathbf{1}_{\{\operatorname{Re} w>c\}}\,e^{g(w)}\,P_{m-1}(w).\end{aligned}} \tag{86}$$

Proof. The recurrence $\partial_w(e^g P_k) = e^g(P_k' + g'P_k)$ with $g'(w) = \sigma_T^2 w - \xi$ gives $\partial_w^k e^g = e^g P_k$ by induction. Since $\partial_{w_*}^{m-1}(w-w_*)^{-1} = (m-1)!\,(w-w_*)^{-m}$ and the Gaussian $|e^g|$ dominates every w_*-derivative of $(w-w_*)^{-1}$ uniformly on the contour for $\{|\operatorname{Re} w_* - c| \ge \epsilon\}$, differentiating (74) under the integral sign $m-1$ times yields (85).

For (86), apply the Leibniz rule to $\frac{1}{2}e^{g(w)}\operatorname{erfc}(z_w)$. The argument $z_w = (\xi - \sigma_T^2 w)/(\sigma_T\sqrt{2})$ is affine with $dz_w/dw = -\sigma_T/\sqrt{2}$, so by the chain rule $\partial_w^j \operatorname{erfc}(z_w) = (-\sigma_T/\sqrt{2})^j \operatorname{erfc}^{(j)}(z_w)$. The $j=0$ term of Leibniz is $\frac{1}{2}\,e^g P_{m-1}\operatorname{erfc}(z_w)$. For $1 \le j \le m-1$, using $\partial_w^{m-1-j}e^g = e^g P_{m-1-j}$ and Lemma 11.3,

$$\begin{aligned}\tfrac{1}{2}\binom{m-1}{j}\big(\partial_w^{m-1-j}e^g\big)\big(\partial_w^j\operatorname{erfc}(z_w)\big) = {} & \tfrac{1}{2}\binom{m-1}{j}e^g P_{m-1-j}\Big(-\tfrac{\sigma_T}{\sqrt{2}}\Big)^j\\ & \times\Big(-\tfrac{2}{\sqrt{\pi}}\Big)(-1)^{j-1}H_{j-1}(z_w)e^{-z_w^2}.\end{aligned} \tag{87}$$

The scalar prefactor simplifies: with $\big(-\frac{\sigma_T}{\sqrt{2}}\big)^j = (-1)^j\big(\frac{\sigma_T}{\sqrt{2}}\big)^j$ the three sign factors give $(-1)^j \cdot (-1)\cdot(-1)^{j-1} = (-1)^{2j} = 1$, while $\frac{1}{2}\cdot\frac{2}{\sqrt{\pi}} = \frac{1}{\sqrt{\pi}}$, so the constant is $\frac{1}{\sqrt{\pi}}\big(\frac{\sigma_T}{\sqrt{2}}\big)^j\binom{m-1}{j}$. The Gaussian factor is parameter-free: with $g(w) = \frac{1}{2}\sigma_T^2 w^2 - w\xi$ and

$$z_w^2 = \frac{(\xi-\sigma_T^2 w)^2}{2\sigma_T^2} = \frac{\xi^2}{2\sigma_T^2} - w\xi + \tfrac{1}{2}\sigma_T^2 w^2, \qquad g(w) - z_w^2 = -\frac{\xi^2}{2\sigma_T^2}, \tag{88}$$

so $e^{g(w)}e^{-z_w^2} = e^{-\xi^2/(2\sigma_T^2)}$, independent of w. This produces the displayed sum. Finally the indicator term differentiates as

$$\partial_w^{m-1}\big(\mathbf{1}_{\{\operatorname{Re} w>c\}}e^g\big) = \mathbf{1}_{\{\operatorname{Re} w>c\}}\,e^g\,P_{m-1}, \tag{89}$$

the indicator being locally constant where $\operatorname{Re} w \neq c$, which completes (86). $\square$

12 Multinomial expansion and the closed-form price

Lemma 12.1 (Cauchy product of pole exponentials)**.** *On $\{w : \min_j |w-\hat{u}_j| \ge \rho > 0\}$,*

$$\prod_{j=1}^{2M}\exp\Big(\frac{B_j}{w-\hat{u}_j}\Big) = \sum_{\mathbf{n}\in\mathbb{Z}_{\ge 0}^{2M}}\prod_{j=1}^{2M}\frac{B_j^{n_j}}{n_j!\,(w-\hat{u}_j)^{n_j}}, \tag{90}$$

absolutely and uniformly, with $\sum_{\mathbf{n}}\big|\prod_j B_j^{n_j}/(n_j!(w-\hat{u}_j)^{n_j})\big| \le \prod_{j=1}^{2M} e^{|B_j|/\rho}$.

Proof. Each factor satisfies $|B_j/(w-\hat{u}_j)| \le |B_j|/\rho$, hence

$$\sum_{n_j\ge 0} \frac{|B_j/(w-\hat{u}_j)|^{n_j}}{n_j!} \le e^{|B_j|/\rho}, \qquad \sum_{n_j\ge 0} \frac{(B_j/(w-\hat{u}_j))^{n_j}}{n_j!} = \exp\Big(\frac{B_j}{w-\hat{u}_j}\Big). \tag{91}$$

The Cauchy product of $2M$ absolutely convergent series converges absolutely to the product of their sums, and the product of the majorants is $\prod_j e^{|B_j|/\rho} < \infty$, uniform on the stated set. $\square$

Lemma 12.2 (Heaviside cover-up for repeated poles)**.** *For pairwise distinct* $w_0, \dots, w_r$ *and* $n_\ell \ge 1$*,*

$$\frac{1}{\prod_{\ell=0}^{r}(w-w_\ell)^{n_\ell}} = \sum_{\ell=0}^{r}\sum_{m=1}^{n_\ell} \frac{R_{\ell,m}}{(w-w_\ell)^m}, \qquad R_{\ell,m} = \frac{1}{(n_\ell-m)!}\,\partial_w^{n_\ell-m}\Big[\prod_{j\ne\ell}(w-w_j)^{-n_j}\Big]_{w=w_\ell}. \tag{92}$$

Proof. Write $F(w) = (w-w_\ell)^{-n_\ell}G_\ell(w)$ with $G_\ell(w) = \prod_{j\ne\ell}(w-w_j)^{-n_j}$, holomorphic and nonzero at w_ℓ. Its Taylor expansion $G_\ell(w) = \sum_{p\ge 0}\frac{G_\ell^{(p)}(w_\ell)}{p!}(w-w_\ell)^p$ multiplied by $(w-w_\ell)^{-n_\ell}$ has, as the coefficient of $(w-w_\ell)^{-m}$ $(1 \le m \le n_\ell)$, the value $G_\ell^{(n_\ell-m)}(w_\ell)/(n_\ell-m)! = R_{\ell,m}$. Summing the principal parts over all poles reconstructs F: the difference is entire and tends to 0 at infinity, hence vanishes by Liouville. $\square$

Theorem 12.3 (Absolutely convergent closed-form call and put)**.** *Under 2.1–2.4, with* $S(\mathbf{n}) = \{j : n_j \ge 1\}$*,*

$$R_\delta^{(\delta,\mathbf{n})} = \prod_{j\in S(\mathbf{n})}\frac{1}{(\delta-\hat{u}_j)^{n_j}}, \tag{93}$$

$$R_{\hat{u}_\ell,m}^{(\delta,\mathbf{n})} = \frac{1}{(n_\ell-m)!}\,\partial_w^{n_\ell-m}\Big[\frac{1}{(w-\delta)\prod_{j\in S(\mathbf{n})\setminus\{\ell\}}(w-\hat{u}_j)^{n_j}}\Big]_{w=\hat{u}_\ell}, \tag{94}$$

$$\Psi_{\mathbf{n}}^{(\delta)} = R_\delta^{(\delta,\mathbf{n})}\,\mathcal{E}(\delta) + \sum_{\ell\in S(\mathbf{n})}\sum_{m=1}^{n_\ell} R_{\hat{u}_\ell,m}^{(\delta,\mathbf{n})}\,\frac{1}{(m-1)!}\,\partial_w^{m-1}\mathcal{E}(w)\big|_{w=\hat{u}_\ell}. \tag{95}$$

Then the European call and put on $S_T = S_0 e^{x_T + rT}$ *with* x_T *characterised by* ϕ_M *of* (58) *are*

$$\boxed{C_M(K,T) = Ke^{-rT}\sum_{\mathbf{n}\in\mathbb{Z}_{\ge 0}^{2M}}\Big(\prod_{j=1}^{2M}\frac{B_j^{n_j}}{n_j!}\Big)\big[\Psi_{\mathbf{n}}^{(1)} - \Psi_{\mathbf{n}}^{(0)}\big], \qquad B_j = iA_j} \tag{96}$$

$$\boxed{\Pi_M(K,T) = C_M(K,T) - \big(S_0 - Ke^{-rT}\big)} \tag{97}$$

where in (96) *the contour data* ($\mathcal{E}$, $\partial^{m-1}\mathcal{E}$) *use the call abscissa* $c \in (1, p^*)$, *and* (97) *is put–call parity.*

Proof. **(1)** By Theorem 10.2, $C_M = \frac{Ke^{-rT}}{2\pi i}\int_{c-i\infty}^{c+i\infty}\phi_M(-iw)e^{-w\tilde{k}}[w(w-1)]^{-1}dw$. **(2)** Lemma 11.1 rewrites the integrand as $e^{g(w)}\prod_j \exp(B_j/(w-\hat{u}_j))\,[w(w-1)]^{-1}$. **(3)** The kernel splits by $1/[w(w-1)] = 1/(w-1) - 1/w$ (Lemma 12.2 on $\{0,1\}$), so

$$C_M = \frac{Ke^{-rT}}{2\pi i}\big[\mathcal{J}^{(1)} - \mathcal{J}^{(0)}\big], \qquad \mathcal{J}^{(\delta)} = \int_{c-i\infty}^{c+i\infty}\frac{e^{g(w)}\prod_j \exp(B_j/(w-\hat{u}_j))}{w-\delta}\,dw, \;\; \delta\in\{0,1\}. \tag{98}$$

(4) On $\operatorname{Re} w = c$, $\rho \geq 1$ by 2.3; Lemma 12.1 expands the product, and the absolute majorant (finite by the Gaussian factor; Theorem 13.1) licenses term-by-term integration (Fubini). **(5)** For each $\mathbf{n}$ the denominator is $(w-\delta)\prod_{j\in S(\mathbf{n})}(w-\hat{u}_j)^{n_j}$ with all poles distinct ($\delta \in \{0,1\}$ off the line $\operatorname{Re} w = c$, and $\operatorname{Re}\hat{u}_j = -\operatorname{Im} u_j$ separated from the line by 2.3). Lemma 12.2 produces (93)–(94); integrating the simple pole at δ by Lemma 11.2 and the order-m pole at $\hat{u}_\ell$ by Lemma 11.4,

$$\frac{\mathcal{J}^{(\delta)}}{2\pi i} = \sum_{\mathbf{n}}\Big(\prod_j \frac{B_j^{n_j}}{n_j!}\Big)\Psi_{\mathbf{n}}^{(\delta)}. \tag{99}$$

(6) Substituting into (98) gives (96). The put (97) is Remark 10.3. □

Corollary 12.4 (Black–Scholes limit; normalisation certificate)**.** *If all $A_j = 0$ (no jump part) and σ_T^2, μ_T are the Black–Scholes variance and the martingale-adjusted mean, then $\Phi_M(u,T) = -\frac{1}{2}\sigma_T^2 u^2 - i\mu_T u$ with $\mu_T = \frac{1}{2}\sigma_T^2$ (from (60) with no poles), and (96) reduces to*

$$C_M(K,T) = S_0\, N(d_1) - Ke^{-rT}\, N(d_2), \qquad d_{1,2} = \frac{\log(S_0/K) + rT}{\sigma_T} \pm \frac{\sigma_T}{2}, \tag{100}$$

with N the standard normal CDF.

Proof. With $A_j = 0$ the multi-index sum reduces to the single term $\mathbf{n} = \mathbf{0}$, where $S(\mathbf{0}) = \varnothing$, $\prod_j B_j^0/0! = 1$, and $\Psi_{\mathbf{0}}^{(\delta)} = R_\delta^{(\delta,\mathbf{0})}\mathcal{E}(\delta) = \mathcal{E}(\delta)$ (empty product $= 1$). Thus $C_M = Ke^{-rT}[\mathcal{E}(1) - \mathcal{E}(0)]$. With both indicators vanishing (any $c > 1$ is admissible in the pole-free case),

$$\mathcal{E}(\delta) = \tfrac{1}{2}e^{g(\delta)}\operatorname{erfc}(z_\delta), \qquad g(\delta) = \tfrac{1}{2}\sigma_T^2\delta^2 - \delta\xi, \quad z_\delta = \frac{\xi - \sigma_T^2\delta}{\sigma_T\sqrt{2}}, \quad \xi = \tilde{k} + \tfrac{1}{2}\sigma_T^2. \tag{101}$$

At $\delta = 0$: $g(0) = 0$, $z_0 = \xi/(\sigma_T\sqrt{2})$, so $\mathcal{E}(0) = \frac{1}{2}\operatorname{erfc}\big(\frac{\xi}{\sigma_T\sqrt{2}}\big) = N\big(-\frac{\xi}{\sigma_T}\big)$ via $N(x) = \frac{1}{2}\operatorname{erfc}(-x/\sqrt{2})$. Now $-\xi/\sigma_T = -(\tilde{k} + \frac{1}{2}\sigma_T^2)/\sigma_T = \big(\log(S_0/K) + rT - \frac{1}{2}\sigma_T^2\big)/\sigma_T = d_2$, since $\tilde{k} = \log(K/S_0) - rT$. Hence $\mathcal{E}(0) = N(d_2)$. At $\delta = 1$: $g(1) = \frac{1}{2}\sigma_T^2 - \xi = \frac{1}{2}\sigma_T^2 - \tilde{k} - \frac{1}{2}\sigma_T^2 = -\tilde{k}$, and $z_1 = (\xi - \sigma_T^2)/(\sigma_T\sqrt{2})$, so

$$\mathcal{E}(1) = \tfrac{1}{2}e^{-\tilde{k}}\operatorname{erfc}\Big(\tfrac{\xi - \sigma_T^2}{\sigma_T\sqrt{2}}\Big) = e^{-\tilde{k}}\, N\Big(-\tfrac{\xi - \sigma_T^2}{\sigma_T}\Big). \tag{102}$$

Here $-(\xi - \sigma_T^2)/\sigma_T = -(\tilde{k} + \frac{1}{2}\sigma_T^2 - \sigma_T^2)/\sigma_T = \big(\log(S_0/K) + rT + \frac{1}{2}\sigma_T^2\big)/\sigma_T = d_1$, and $e^{-\tilde{k}} = \frac{S_0}{K}e^{rT}$ from (3), so

$$Ke^{-rT}\mathcal{E}(1) = Ke^{-rT}\cdot\tfrac{S_0}{K}e^{rT}N(d_1) = S_0 N(d_1).$$

Therefore

$$C_M = Ke^{-rT}\mathcal{E}(1) - Ke^{-rT}\mathcal{E}(0) = S_0N(d_1) - Ke^{-rT}N(d_2),$$

which is (100). The exact recovery of Black–Scholes fixes $\xi = \tilde{k} + \mu_T$, the image $\hat{u}_j = iu_j$, $B_j = iA_j$, and the prefactor Ke^{-rT} as in (96). □

13 Absolute convergence for all admissible parameters

Theorem 13.1 (Unconditional absolute convergence)**.** *Under 2.1–2.4 the series (96) converges absolutely, with the a priori majorant*

$$\sum_{\mathbf{n}}\Big|\prod_j \frac{B_j^{n_j}}{n_j!}\Big|\big(|\Psi_{\mathbf{n}}^{(1)}| + |\Psi_{\mathbf{n}}^{(0)}|\big) \;\leq\; \frac{Ke^{-rT}}{\sigma_T\sqrt{2\pi}}\cdot\frac{(2c-1)\, e^{\sigma_T^2c^2/2 - c\xi}}{c(c-1)}\cdot\prod_{j=1}^{2M} e^{|A_j|/\rho} \;<\; \infty. \tag{103}$$

Consequently $C_M(K,T)$ and $\Pi_M(K,T)$ are well defined for every admissible parameter set, and $C_M \to C$ at the Stahl rate of Theorem 8.1.

Proof. Estimate at the level of the contour integral, before partial fractions, so that the multi-pole bookkeeping never enters. For $\delta \in \{0,1\}$ and $w = c + it$, Lemma 12.1 bounds the multinomial sum:

$$\Big|\sum_{\mathbf{n}} \prod_j \frac{B_j^{n_j}}{n_j!(w-\hat{u}_j)^{n_j}}\Big| \le \sum_{\mathbf{n}} \prod_j \frac{(|B_j|/\rho)^{n_j}}{n_j!} = \prod_{j=1}^{2M} e^{|B_j|/\rho}, \tag{104}$$

while

$$\Big|\frac{e^{g(w)}}{w-\delta}\Big| = \frac{e^{\sigma_T^2(c^2-t^2)/2-c\xi}}{\sqrt{(c-\delta)^2+t^2}}. \tag{105}$$

Therefore, integrating along the line and using $\sqrt{(c-\delta)^2+t^2} \ge |c-\delta|$ with the Gaussian integral $\int_{\mathbb{R}} e^{-\sigma_T^2 t^2/2}\,dt = \sqrt{2\pi/\sigma_T^2}$,

$$\frac{1}{2\pi}\int_{\mathbb{R}}\Big|\frac{e^{g(c+it)}}{c-\delta+it}\Big| \prod_j e^{|B_j|/\rho}\,dt \le \frac{\prod_j e^{|B_j|/\rho}}{\sigma_T\sqrt{2\pi}} \cdot \frac{e^{\sigma_T^2 c^2/2-c\xi}}{|c-\delta|}. \tag{106}$$

By steps (4)–(5) of Theorem 12.3 this dominates $\sum_{\mathbf{n}} |\prod_j B_j^{n_j}/n_j!|\,|\Psi_{\mathbf{n}}^{(\delta)}|$ (the term-by-term integral is bounded by the integral of the absolute majorant). Multiply by Ke^{-rT}, then add $\delta = 1$ ($|c-\delta| = c-1$, since $c > 1$) and $\delta = 0$ ($|c-\delta| = c$): $\frac{1}{c-1}+\frac{1}{c} = \frac{2c-1}{c(c-1)}$, and $\prod_j e^{|B_j|/\rho} = \prod_j e^{|A_j|/\rho}$, giving (103). Finiteness uses only $\sigma_T^2 > 0$ (2.2) and $\rho > 0$ (2.3); there is no condition on the magnitudes of the A_j, hence the bound holds for every admissible parameter set. The limit statement is Theorem 8.1 together with local Lipschitz continuity of exp on the contour: $\Phi_M \to \Phi$ uniformly on $\operatorname{Re} w = c$ implies $\phi_M \to \phi$ uniformly there, so $C_M \to C$ at the same geometric rate. The put converges by (97). □

Corollary 13.2 (Quadrature-free pipeline)**.** *$C_M(K,T)$ and $\Pi_M(K,T)$ are computed by:* (i) *the Müntz–Tau coefficients $a_k(v)$ and the lattice $d_k(v)$ for $k \le 2M$ (Theorems 4.1, 5.2);* (ii) *the Chebyshev recurrence (Theorem 6.3), at $O(M^2)$;* (iii) *root-finding of $Q_M(T^\mu;\cdot)$ to obtain $\{u_j, A_j, \sigma_T, \mu_T\}$, at $O(M^2)$ per T (Theorem 7.1, Lemma 9.3);* (iv) *evaluation of* (96) *via the P_k recurrence and the* erfc*/Hermite kernel* (86)*, absolutely convergent with the $\prod_j e^{|A_j|/\rho}$ control of Theorem 13.1. No Fourier inversion, no cosine series, no truncation interval, and no special function beyond* erfc *are used.*

14 Summary

T1 (resummation). The cgf $\Phi(v,T) = T\,\widetilde{\Phi}(T^\mu; v)$ (Theorem 5.2) is resummed by its diagonal $[M/M]$ Padé approximant built on the lattice coefficients $\{d_k(v)\}$ through the Chebyshev recurrence (Theorem 6.3) in $O(M^2)$ ring operations, the denominator being the reversed orthogonal polynomial (Theorem 7.1); the rate as $M \to \infty$ is Stahl's (Theorem 8.1), with the a-posteriori control of Theorem 8.2.

T2 (price). The European call is the series (96) (Theorem 12.3) in the partial-fraction data $(\sigma_T, \mu_T, \{u_j, A_j\})$, with kernel $\mathcal{E} = \frac{1}{2}e^g \operatorname{erfc} -\mathbf{1}\, e^g$ and every derivative the closed Hermite form (86), derived from the Parseval/Lewis representation (Theorem 10.2; payoff transforms

Lemma 10.1), the rotation Lemma 11.1 ($B_j = iA_j$, $\hat{u}_j = iu_j$, $\xi = \tilde{k} + \mu_T$), the Cauchy product Lemma 12.1, and the Schwinger identity Lemma 11.2; the put is (97) by parity (Remark 10.3).

T3 (convergence and limit). The series is absolutely convergent for every admissible parameter set, dominated by the right-hand side of (103) (Theorem 13.1), under the sole structural requirement $\sigma_T^2 > 0$; the pole-free specialisation returns Black–Scholes exactly (Corollary 12.4), certifying all normalisations, and $C_M \to C$ at the Stahl rate.

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